GUESS THE NUMBER GAME

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ABSTRACT

The “Guess The Number” project is a simple console-based game developed using the C programming language. In this game, the computer generates a random number between 1 and 100, and the player has to guess the correct number within 5 attempts. After every guess, the program gives hints whether the guessed number is too high or too low. The project demonstrates the use of loops, conditional statements, random number generation, and user interaction in C programming. This project is useful for beginners to understand basic programming concepts and game logic implementation.

INTRODUCTION

Guess The Number is a mini-game application written in C language. The main objective of this project is to improve logical thinking and programming skills. The game creates excitement for users by generating a random number and giving limited chances to guess it correctly.

The project uses:

Random number generation

Conditional statements

Loops

User input handling

This project is beginner-friendly and helps students understand the fundamentals of C programming.

ALGORITHM

Step1: Start the program.
Step2: Generate a random number (1–100).
Step3: Give user 5 attempts.
Step4: Read user guess.
Step5: Compare guess with number.
Step6: Display “Too High” or “Too Low”.
Step7: If correct → Display success message.
Step8: If attempts end → Display Game Over.
Step9: Stop.

FLOWCHART

```
START
  |
Generate Random Number
  |
Enter Guess
  |
Guess Correct?
 /      \
Yes      No
 |        |
Win     High/Low Hint
 |        |
Stop   Attempts Left?
        /   \
      Yes    No
       |      |
    Enter Again Game Over
                |
               Stop
```

METHODOLOGY

The project follows a simple gaming methodology:
1.The system generates a random number using rand() function.
2.The user enters guesses through keyboard input.
3.The program checks the guess using conditional statements.
4.Feedback is provided after every attempt.
5.The loop continues until:
The user guesses correctly, or
All 5 attempts are completed.
6.The final result is displayed.

RESULT

main.c Output

```
=== Guess The Number Game ===
Guess a number between 1 and 100
You have only 5 attempts.

Enter your guess: 23
Too low!
Attempts remaining: 4

Enter your guess: 65
Too low!
Attempts remaining: 3

Enter your guess: 89
Too high!
Attempts remaining: 2

Enter your guess: 76
Too low!
Attempts remaining: 1

Enter your guess: 80
Too high!
Attempts remaining: 0

Game Over! You used all 5 attempts.
The correct number was 79
```

main.c Output

```
=== Guess The Number Game ===
Guess a number between 1 and 100
You have only 5 attempts.

Enter your guess: 34
Too high!
Attempts remaining: 4

Enter your guess: 21
Too high!
Attempts remaining: 3

Enter your guess: 11
Congratulations! You guessed the number in 3
    attempts.


=== Code Execution Successful ===
```

main.c Output

```
=== Guess The Number Game ===
Guess a number between 1 and 100
You have only 5 attempts.

Enter your guess: 63
Too high!
Attempts remaining: 4

Enter your guess: 32
Too high!
Attempts remaining: 3

Enter your guess: 12
Congratulations! You guessed the number in 3
    attempts.


=== Code Execution Successful ===
```

CONCLUSION

The Guess The Number project successfully demonstrates the implementation of a simple game using C programming. The project helps in understanding loops, conditional statements, random number generation, and user interaction. It is an excellent beginner-level project for improving logical and programming skills.

FUTURE WORK

1.Add difficulty levels (Easy/Medium/Hard)
2.Add score calculation system
3.Store high scores in a file
Add multiplayer mode
4.Create a graphical user interface (GUI)
5.Add timer-based gameplay